

① How many elements does each of these sets have?

a) $P(\{\{a, b\}, \{a, b\}\}) - 3; 2^3 = 8$

b) $P(\{\emptyset, a, \{a\}, \{\{a\}\}\}) - 4; 2^4 = 16$

c) $P(P(\emptyset)) - 0; 2^0 = 1; 2^1 = 2$

② How many different elements does A^n have when A has m elements and n is a positive int.

$$A \cdot A = m^2 \Rightarrow A^2 = m^2 \Rightarrow A^n = m^n$$

③ a) $P(x): x^2 < 3$

$$x \in \{-1, 0, 1\}$$

$$b) Q(x) : x^2 > x \Rightarrow$$

$$\Rightarrow x \in (-\infty; -1] \cup [2; +\infty)$$

$$c) R(x) : 2x + 1 = 0$$

$$x = -0,5 \text{ — isn't int}$$

$$\Rightarrow \{x \in \mathbb{Z} \mid 2x + 1 = 0\} = \emptyset$$

⑤ If A, B and C are sets, then

$$\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$$

$$x \in \overline{A \cap B \cap C}$$

$$x \notin A \cap B \cap C \Rightarrow x \notin A; x \notin B; x \notin C$$

$$x \in \overline{A}; x \in \overline{B}; x \in \overline{C}$$

$$x \in \overline{A} \cup \overline{B} \cup \overline{C} \text{ and } x \in \overline{A \cap B \cap C}$$

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$A \wedge B \wedge C$	$\overline{A \wedge B \wedge C}$	$\bar{A} \vee \bar{B} \vee \bar{C}$
1	1	1	0	0	0	1	0	0
1	1	0	0	0	1	0	1	1
1	0	1	0	1	0	0	1	1
1	0	0	0	1	1	0	1	1
0	1	1	1	0	0	0	1	1
0	1	0	1	0	1	0	1	1
0	0	1	1	1	0	0	1	1
0	0	0	1	1	1	0	1	1

